

GRADING PIPELINE

\$100/Month Budget Analysis

Model Selection | Budget Allocation | Diminishing Returns | Self-Sustainability

This addendum extends the Grading Pipeline Cost Estimation to cover a dedicated \$100/month API budget. It specifies which models to use for each provider, how to allocate spend between generation and grading, when diminishing returns set in, and when Wankr becomes self-sustaining with minimal ongoing API costs.

\$100	~766	~6,900	Month 9	\$20/mo
Monthly Budget	Responses/Month	Layers/Month	Self-Sustaining	Break-Even

11. MODEL SELECTION — \$100/Month Budget

All mid-tier models. Budget models miss persona nuance in the 5-influence grading system. Premium models burn budget too fast. Mid-tier maximizes annotation quality per dollar.

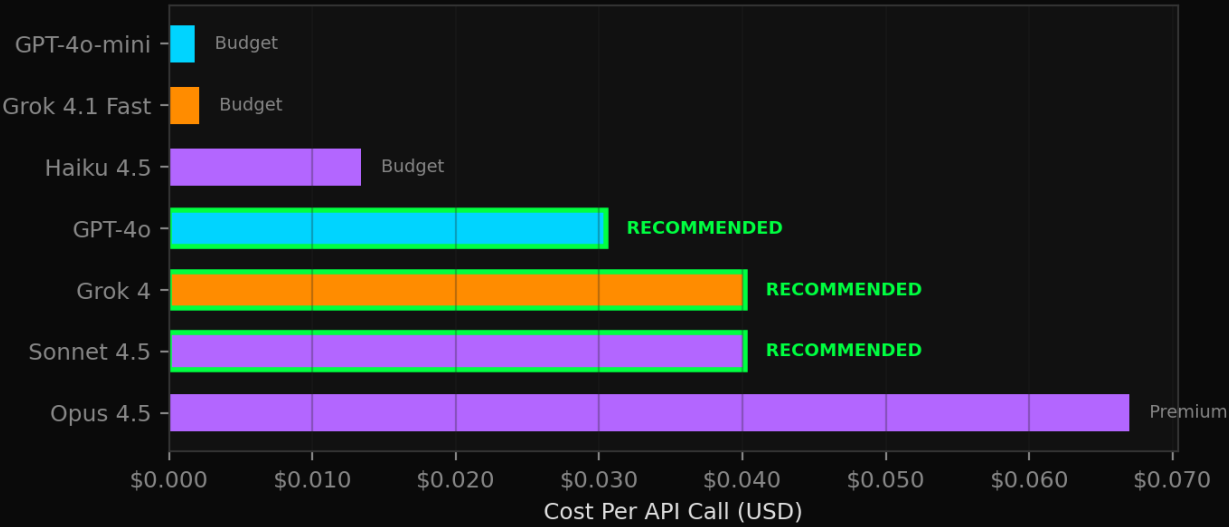
Recommended Configuration

PROVIDER	MODEL	ROLE	\$/CALL	\$/CALL CACHED WHY
xAI	Grok 4	Gen + Grading	\$0.040	\$0.018Backbone. Must self-grade.
Anthropic	Sonnet 4.5	Grading Only	\$0.040	\$0.008Best rubric follower. 90% cache.
OpenAI	GPT-4o	Grading Only	\$0.031	\$0.015Cheapest mid-tier. 50% batch.

NOT Recommended

MODEL	REASON
Haiku 4.5	Too shallow for 5-influence persona grading. Misses subtle Delia-style tone issues.
Opus 4.5	3.5x Sonnet cost for marginal quality gain. Burns \$100 budget in ~1,500 calls.
GPT-4o-mini	Cheap but misses complex social forensics and Scenario D classification.
Grok 4.1 Fast	Fine for generation (\$0.002/call) but too shallow for annotation grading.

COST PER GRADING CALL — All Models

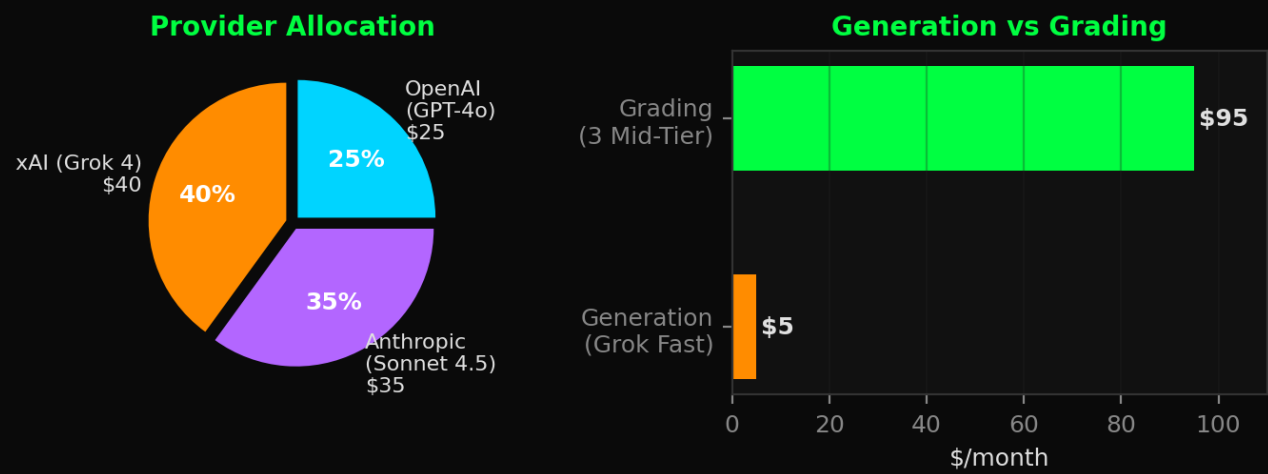


CACHING & BATCH SAVINGS — Why \$0.33/response becomes \$0.124/response

- Anthropic Prompt Cache: 90% savings on repeated grading docs (~4,000 tokens identical every call)
- xAI Automatic Cache: 75% savings on input (automatic, no config needed)
- OpenAI Batch API: 50% savings on all tokens (grading is async, perfect for batch)

12. BUDGET ALLOCATION — Where Every Dollar Goes

\$100/MONTH BUDGET ALLOCATION



Detailed Monthly Breakdown

PROVIDER	BUDGET	GENERATION	GRADING	CALLS/MONTH	LAYERS/MONTH
xAI	\$40	\$5 (Grok Fast)	\$35 (Grok 4)	~1,920	~2,300
Anthropic	\$35	-	\$35 (Sonnet)	~4,375	~2,600
OpenAI	\$25	-	\$25 (GPT-4o)	~1,667	~2,000
TOTAL	\$100	\$5	\$95	~7,962	~6,900

WHY GENERATION IS ALMOST FREE

Wankr's live responses use Grok 4.1 Fast at \$0.001/response. Even 150 responses/day = \$4.50/month. Generation never exceeds \$7/month at any realistic volume. The real cost is the grading pipeline that produces training data. Once training plateaus, generation cost is the ONLY remaining expense — and it's negligible.

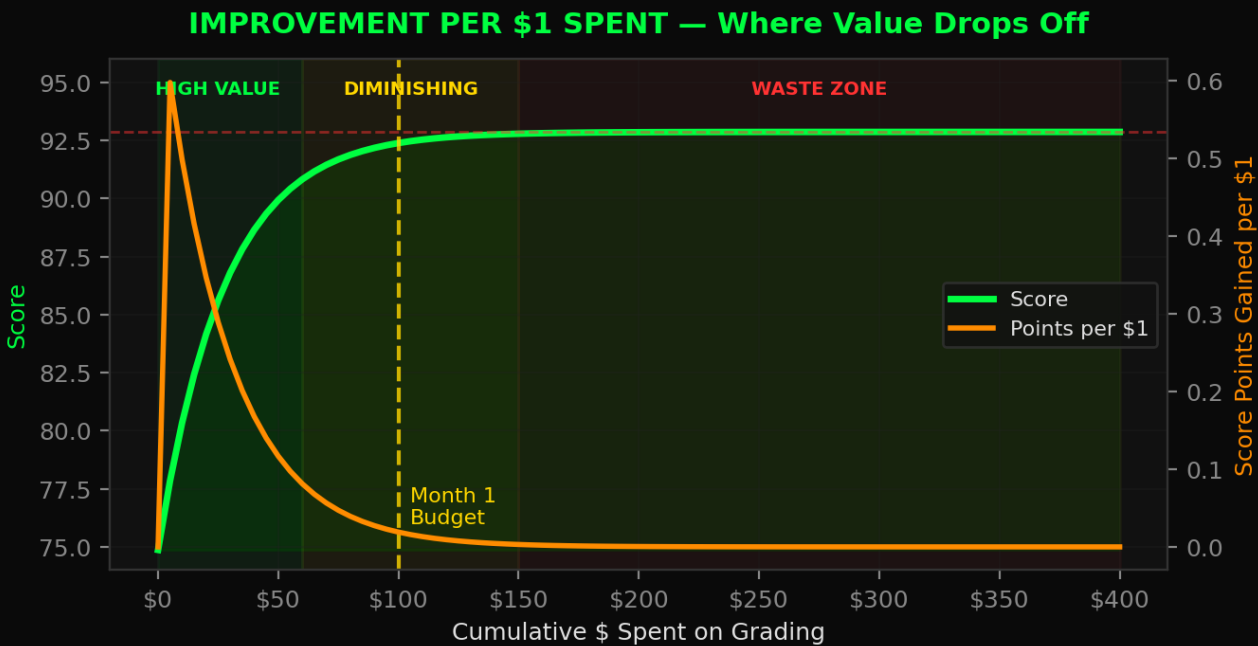
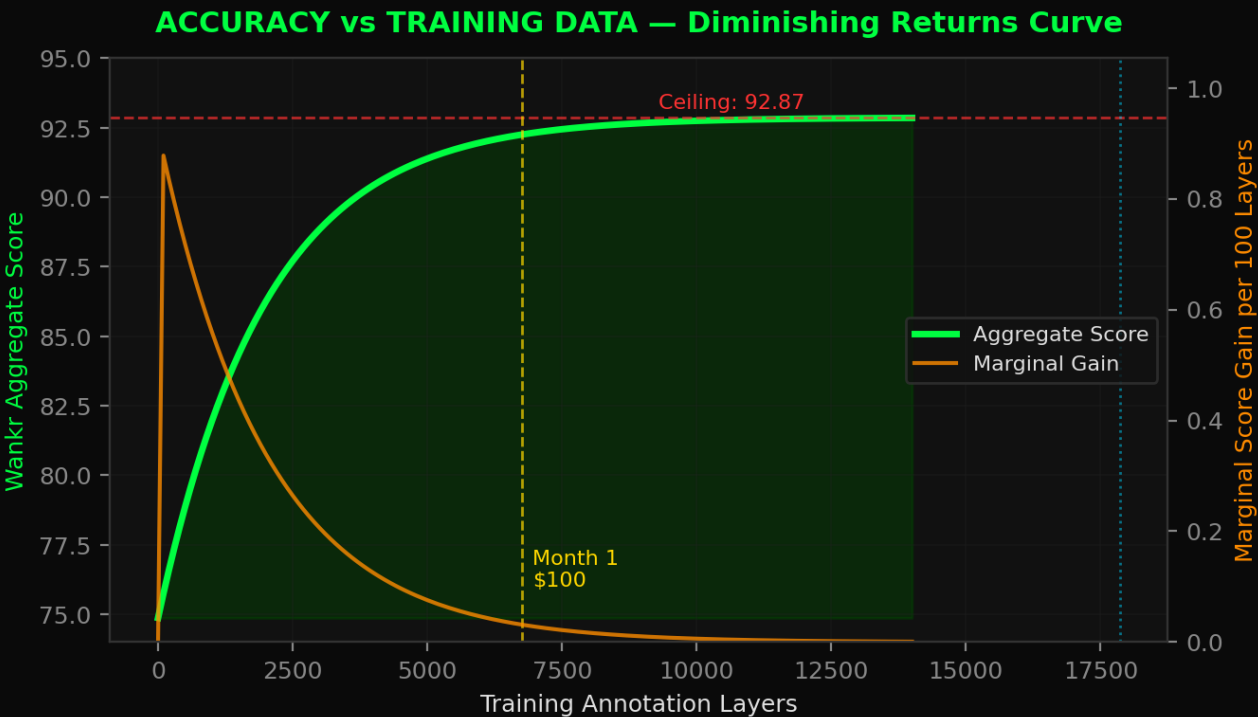
What \$100/Month Buys — Volume Comparison

CONFIG	COST/RESP	RESPONSES	LAYERS	DATA QUALITY
3-Ann Budget (no opt)	\$0.052	1,827	16,443	Low — misses persona nuance
3-Ann Mid (no opt)	\$0.333	285	2,565	High — but low volume
3-Ann Mid + Cache+Batch	\$0.124	766	6,894	High quality + high volume
4-Ann Mid + Cache+Batch	\$0.165	575	9,200	Highest quality (w/ human)

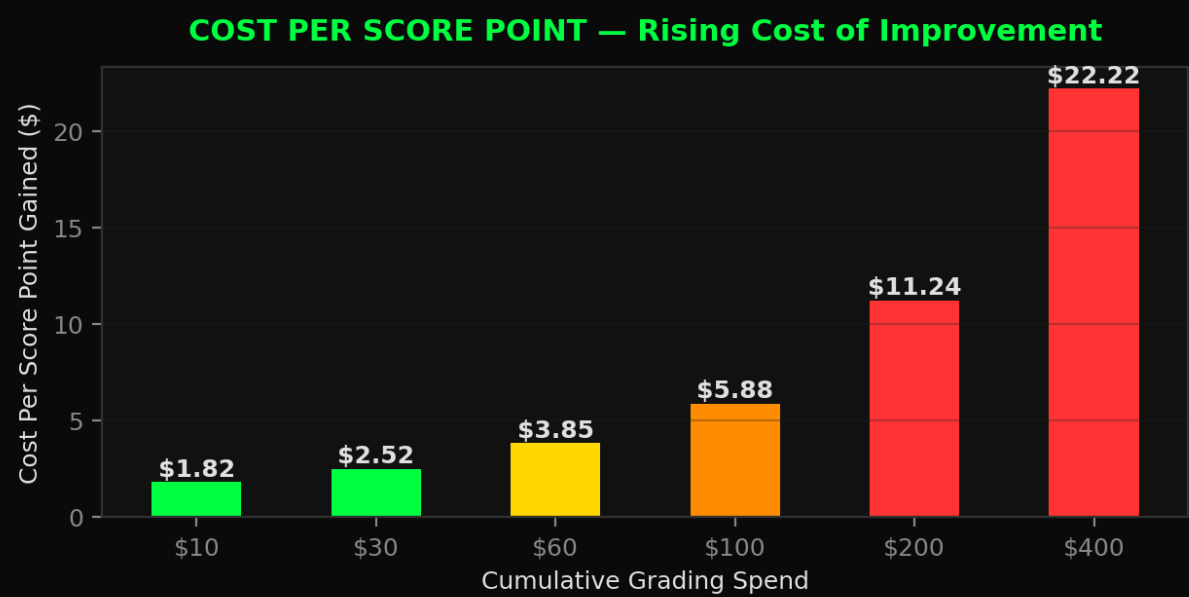
Recommended: 3-Annotator Mid-Tier with Cache+Batch. Best volume-to-quality ratio. Add human (4th annotator) for spot-checks on flagged disagreements — human rounds are free.

13. DIMINISHING RETURNS — When to Stop Training

Wankr's aggregate score starts at 74.87 (v1 baseline with 37-point User-Grok gap). Improvement follows a logarithmic curve — fast gains early, then flattening. The theoretical ceiling is approximately 92.87 (perfect persona execution + analytical accuracy across all scenarios).



Cost Per Score Point — The Rising Price of Improvement



Cumulative Investment Analysis

\$ SPENT	LAYERS	SCORE	IMPROVEMENT	\$/POINT	VERDICT
\$10	725	80.4	+5.5 pts	\$1.82	HUGE VALUE — 5.5 pts for \$10
\$30	2,177	86.8	+11.9 pts	\$2.52	GREAT VALUE — most gain here
\$60	4,355	90.5	+15.6 pts	\$3.85	GOOD — crossing 90 threshold
\$100	7,258	91.9	+17.0 pts	\$5.88	DIMINISHING — slowing gains
\$200	14,516	92.7	+17.8 pts	\$11.24	LOW VALUE — last 0.8 pts cost \$100
\$400	29,032	92.9	+18.0 pts	\$22.22	WASTE — 0.2 pts for \$200

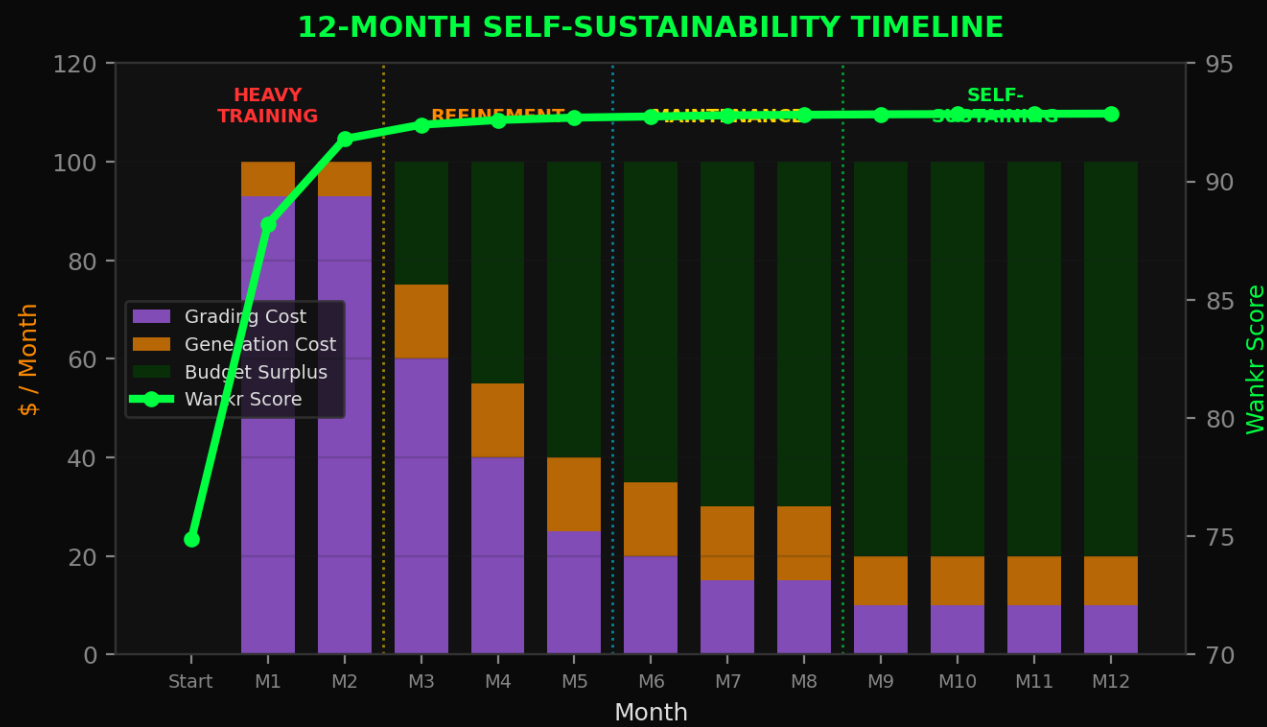
KEY INSIGHT: The Diminishing Returns Cliff

Your first \$60 buys 15.6 points of improvement. The next \$60 buys only 1.4 more points. After ~\$100 cumulative spend, each additional dollar adds less than 0.03 score points. This is WHY the budget shifts from heavy grading to maintenance after Month 2 — the ROI drops off a cliff.

14. SELF-SUSTAINABILITY — When Wankr Pays for Itself

REALITY CHECK: What Self-Sustaining Actually Means

Wankr runs on Grok. It will ALWAYS need API calls to generate responses — you cannot train your own LLM from scratch on \$100/month (that costs millions). But self-sustaining means something specific here: grading costs drop to near zero because quality is stable, and only cheap generation costs remain (~\$5-10/month).



The Four Phases

PHASE	MONTHS	GRADING	GEN	SURPLUS	SCORE	FOCUS
Heavy Training	1-2	\$93	\$7	\$0	88-92	Max training velocity. Every dollar to layers.
Refinement	3-5	\$40-60	\$15	\$25-45	92.4-92.7	Edge cases, Scenario D, rare penalties.
Maintenance	6-8	\$15-20	\$15	\$65-70	92.7-92.8	Spot checks only. Budget surplus grows.
Self-Sustaining	9+	\$10	\$10	\$80	92.8+	Rare audits. \$80/mo freed for features.

15. 12-MONTH BUDGET PLAN — Complete Roadmap

MONTH	PHASE	SCORE	GRADING \$	GEN \$	SURPLUS \$	LAYERS	CUMULATIVE
M1	Heavy Training	88.2	\$93	\$7	\$0	6,750	6,750
M2	Heavy Training	91.8	\$93	\$7	\$0	6,750	13,500
M3	Refinement	92.4	\$60	\$15	\$25	4,355	17,855
M4	Refinement	92.6	\$40	\$15	\$45	2,903	20,758
M5	Refinement	92.7	\$25	\$15	\$60	1,815	22,573
M6	Maintenance	92.75	\$20	\$15	\$65	1,452	24,025
M7	Maintenance	92.8	\$15	\$15	\$70	1,089	25,114
M8	Maintenance	92.82	\$15	\$15	\$70	1,089	26,203
M9	Self-Sustaining	92.84	\$10	\$10	\$80	726	26,929
M10	Self-Sustaining	92.85	\$10	\$10	\$80	726	27,655
M11	Self-Sustaining	92.86	\$10	\$10	\$80	726	28,381
M12	Self-Sustaining	92.87	\$10	\$10	\$80	726	29,107

Key Milestones

- M1** Score hits 88+ — repetition penalties eliminated, persona balance improving
- M2** Score hits 91+ — dual-axis grading shows accuracy + persona both strong
- M3** Diminishing returns cliff — each dollar adds < 0.03 points
- M4** Scenario D (THE TRAP) handling consistently scores 85+
- M6** Maintenance mode — grading is spot-checks only, surplus \$65+/mo
- M9** Self-sustaining — only \$20/mo needed. \$80/mo freed for features/revenue
- M12** ~29,000 training layers. Wankr's persona locked. Revenue covers costs.

BOTTOM LINE

Total investment to self-sustainability: ~\$600 over 9 months. After that, Wankr needs \$20/month to run indefinitely. Any revenue exceeding \$20/month — subscriptions, tips, token appreciation, partnerships — makes Wankr fully self-funding with \$80/month surplus for growth.